

Topic I: System Theory

A **system** is a collection of components working together towards a common goal. The main of a system is to receive inputs and transform these into outputs. A **closed** system is a type of system that does not interact with its environment. An **open** system is a type of system that interacts with its environment.

Organisations are not isolated entities, they are open systems. An organisation is seen primarily as an activity system (or more accurately, a collection of interacting activity systems) that is affected by forces in its environment. This means that the success of any organisation will depend on how well it integrates with aspects of its environment.

A system exists and functions within an environment containing other systems. If a system is one of the components of a larger system, it is a **subsystem**, and the larger system is its environment. A **subsystem** is a component of a large system in a system environment. A business is an example of an organizational system in which economic resources (input) are transformed by various business processes (processing) into goods and services (output). Information systems provide information (feedback) about the operations of the system to management for the direction and maintenance of the system (control) as it exchanges inputs and outputs with its environment.

Characteristics of a System

Some characteristics are present in all systems including organization (order), interaction, interdependence, integration and central objective.

1. Organization

Organization implies structure and order. It is the arrangement of components that helps to achieve objectives. In the design of a business system, for example, the hierarchical relationships starting with the president on top and leading downward to the blue – collar workers represents the organization structure. Such an arrangement portrays a system – subsystem relationship,

defines the authority structure, specifies the formal flow of communication and formalizes the chain of command. Like – wise, a computer system is designed around an input device, a central processing unit, an output device and one or more storage units. When linked together they work as a whole system for producing information.

2. Interaction

Interaction refers to the manner in which each component functions with other components of the system. In an organization, for example, purchasing must interact with production, advertising with sales and payroll with personnel. In a computer system, the central processing unit must interact with the input device to solve a problem. In turn, the main memory holds programs and data that the arithmetic unit uses for computation. The interrelationship between these components enables the computer to perform.

3. Interdependence

Interdependence means that parts of the organization or computer system depend on one another. They are coordinated and linked together according to a plan. One subsystem depends on the input of another subsystem for proper functioning: that is, the output of one subsystem is the required input for another subsystem. This interdependence is crucial in systems work.

An integrated information system is designed to serve the needs of authorized users (department heads, managers, etc.) for quick access and retrieval via remote terminals. The interdependence between the personnel subsystem and the organization's users is obvious.

In summary, no subsystem can function in isolation because it is dependent on the data (inputs) it receives from other subsystems to perform its required tasks. Interdependence is further illustrated by the activities and support of systems analysts, programmers, and the operations staff in a computer centre. A decision to computerize an application is initiated by the user, analyzed and designed by the analyst, programmed and tested by the programmer, and run by

the computer operator. None of these persons can perform property without the required input from others in the computer center subsystem.

4. Integration

Integration refers to the holism of systems. Synthesis follows analysis to achieve the central objective of the organization. Integration is concerned with how a system is tied together. It is more than sharing a physical part or location. It means that parts of the system work together within the system even though each part performs a unique function. Successful integration will typically produce a synergistic effect and greater total impact than if each component works separately.

5. Central Objective

The last characteristic of a system is its central objective. Objectives may be real or stated. Although a stated objective may be the real objective, it is not uncommon for an organization to state one objective and operate to achieve another. The important point is that users must know the central objective of a computer application early in the analysis for a successful design and conversion. Political as well as organizational considerations often cloud the real objective. This means that the analyst must work around such obstacles to identify the real objective of the proposed change.

Formal Systems

A formal system is a system that is designed and developed using a set of well-established organizational policies, procedures, and principles, guidelines to coordinate and facilitate communication between different functional units and the processes they support, and to meet the overall information needs of the business.

Informal Systems

An informal system is a system created by ad hoc, informal work groups to support information needs that cannot be met by formal systems. These are powerful systems that meet unique needs and thrive in many organizations.

Informal systems do not follow any formal or pre-established rules for collecting, processing, storing, or disseminating data.